EXPERIMENT – 3

AIM: To design and implement an Employee–Department–Project database schema and perform SQL queries using Selection, Projection, Aggregate Functions, GROUP BY, HAVING, CASE expressions, and ORDER BY.

SOURCE CODE:

DATABASE CREATING

```
CREATE DATABASE EmployeeManagement;
USE EmployeeManagement;
```

TABLE CREATING

```
CREATE TABLE Department (
        dept_id INT PRIMARY KEY,
        dept_name VARCHAR(50) NOT NULL,
        location VARCHAR(50) );

CREATE TABLE Employee (
        emp_id INT PRIMARY KEY,
        emp_name VARCHAR(50) NOT NULL,
        gender CHAR(1),  salary DECIMAL(10,2),
        hire_date DATE,
        dept_id INT,
        FOREIGN KEY (dept_id)
                REFERENCES Department(dept_id)
);

CREATE TABLE Project (
        project_id INT PRIMARY KEY,
        project_name VARCHAR(100) NOT NULL,
        budget DECIMAL(12,2),
        dept_id INT,
        FOREIGN KEY (dept_id)
                REFERENCES Department(dept_id)
);

CREATE TABLE Employee_Project (
        emp_id INT,
        project_id INT,
        hours_worked INT,
        PRIMARY KEY (emp_id, project_id),
        FOREIGN KEY (emp_id)
                REFERENCES Employee(emp_id),
```

```
        FOREIGN KEY (project_id)
            REFERENCES Project(project_id)
);
```

INSERTING DEPARTMENTS

```
INSERT INTO Department (dept_id, dept_name, location)
VALUES
(1, 'Development', 'Delhi'),
(2, 'Human Resources', 'Mumbai'),
(3, 'Accounting', 'Bangalore'),
(4, 'Sales', 'Pune'),
(5, 'Innovation', 'Hyderabad');
```

INSERTING 30 EMPLOYEES

```
INSERT INTO Employee
(emp_id, emp_name, gender, salary, hire_date, dept_id)
VALUES
(101, 'Aarush Khanna', 'M', 78000, '2021-01-15', 1),
(102, 'Ishita Roy', 'F', 69000, '2022-03-10', 1),
(103, 'Vivaan Sinha', 'M', 84000, '2020-07-20', 1),
(104, 'Myra Joshi', 'F', 74000, '2021-11-05', 1),
(105, 'Reyansh Kapoor', 'M', 97000, '2019-06-18', 1),
(106, 'Anika Bhatia', 'F', 66000, '2023-01-12', 1),
(107, 'Kabir Malhotra', 'M', 57000, '2022-05-15', 2),
(108, 'Aadhya Menon', 'F', 64000, '2021-09-22', 2),
(109, 'Vihaan Arora', 'M', 60000, '2020-04-11', 2),
(110, 'Kiara Saxena', 'F', 69000, '2023-02-19', 2),
(111, 'Advait Rao', 'M', 62000, '2022-08-01', 2),
(112, 'Saanvi Desai', 'F', 73000, '2019-12-10', 2),
(113, 'Dhruv Mehta', 'M', 90000, '2020-01-25', 3),
(114, 'Navya Pillai', 'F', 94000, '2019-08-14', 3),
(115, 'Arnav Kulkarni', 'M', 78000, '2021-06-30', 3),
(116, 'Tara Chawla', 'F', 87000, '2022-10-17', 3),
(117, 'Ishaan Nair', 'M', 75000, '2023-03-21', 3),
(118, 'Riddhi Shah', 'F', 92000, '2020-11-09', 3),
(119, 'Krish Bansal', 'M', 72000, '2021-02-28', 4),
(120, 'Avni Khurana', 'F', 68000, '2022-04-15', 4),
(121, 'Rudra Mishra', 'M', 76000, '2020-09-12', 4),
(122, 'Pari Anand', 'F', 83000, '2019-07-07', 4),
(123, 'Yuvaan Sethi', 'M', 71000, '2023-01-05', 4),
(124, 'Diya Oberoi', 'F', 79000, '2021-12-18', 4),
(125, 'Atharv Reddy', 'M', 99000, '2018-05-20', 5),
(126, 'Meher Kaur', 'F', 93000, '2020-03-16', 5),
(127, 'Shaurya Das', 'M', 89000, '2021-10-08', 5),
(128, 'Zoya Fernandes', 'F', 95000, '2019-11-25', 5),
```

(129, 'Ayaan Thakur', 'M', 81000, '2022-07-14', 5),
(130, 'Vanya Kapoor', 'F', 98000, '2023-02-02', 5);

INSERTING 8 PROJECTS

INSERT INTO Project (project_id, project_name, budget, dept_id) VALUES
(201, 'Smart Web Portal', 550000, 1),
(202, 'Cloud Infrastructure', 800000, 1),
(203, 'Staff Development Program', 300000, 2),
(204, 'Budget Forecasting System', 650000, 3),
(205, 'Online Sales Campaign', 480000, 4),
(206, 'Customer Insights Study', 400000, 4),
(207, 'Predictive Analytics Platform', 950000, 5),
(208, 'Cybersecurity Framework', 1100000, 5);

ASSIGNING EMPLOYEES TO PROJECTS

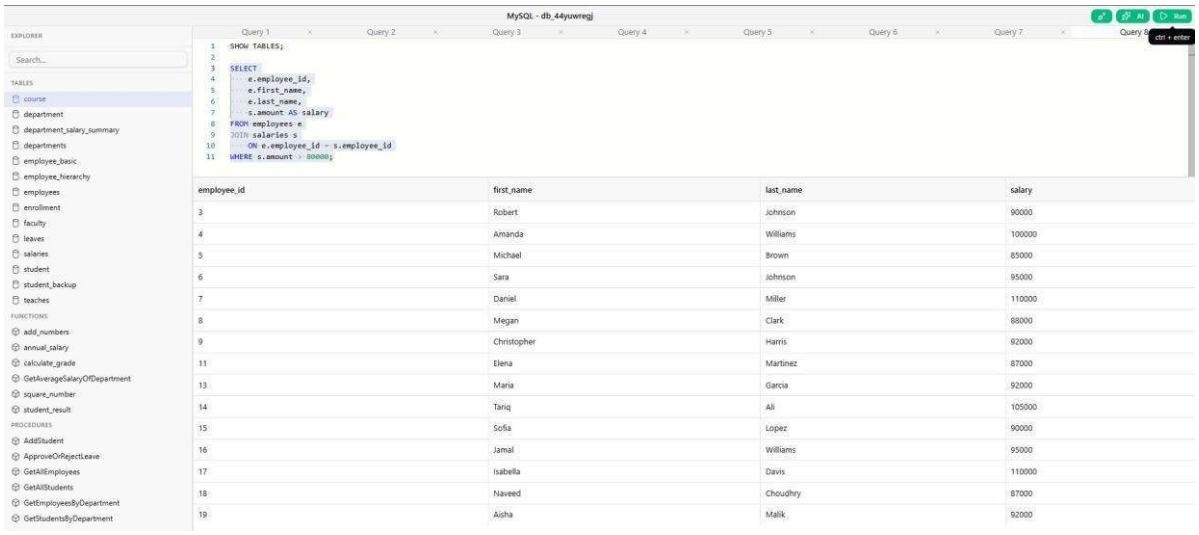
INSERT INTO Employee_Project (emp_id, project_id, hours_worked)
VALUES
(101, 201, 120),
(102, 201, 100),
(103, 202, 150),
(104, 202, 110),
(105, 201, 130),
(106, 202, 90),
(107, 203, 100),
(108, 203, 120),
(109, 203, 80),
(110, 203, 110),
(111, 203, 90),
(112, 203, 130),
(113, 204, 140),
(114, 204, 150),
(115, 204, 120),
(116, 204, 135),
(117, 204, 100),
(118, 204, 145),
(119, 205, 110),
(120, 205, 100),
(121, 206, 120),
(122, 205, 140),
(123, 206, 90),
(124, 206, 130),
(125, 207, 160),
(126, 207, 140),
(127, 208, 150),

(128, 208, 160),
(129, 207, 120),
(130, 208, 135);

SQL QUERIES:

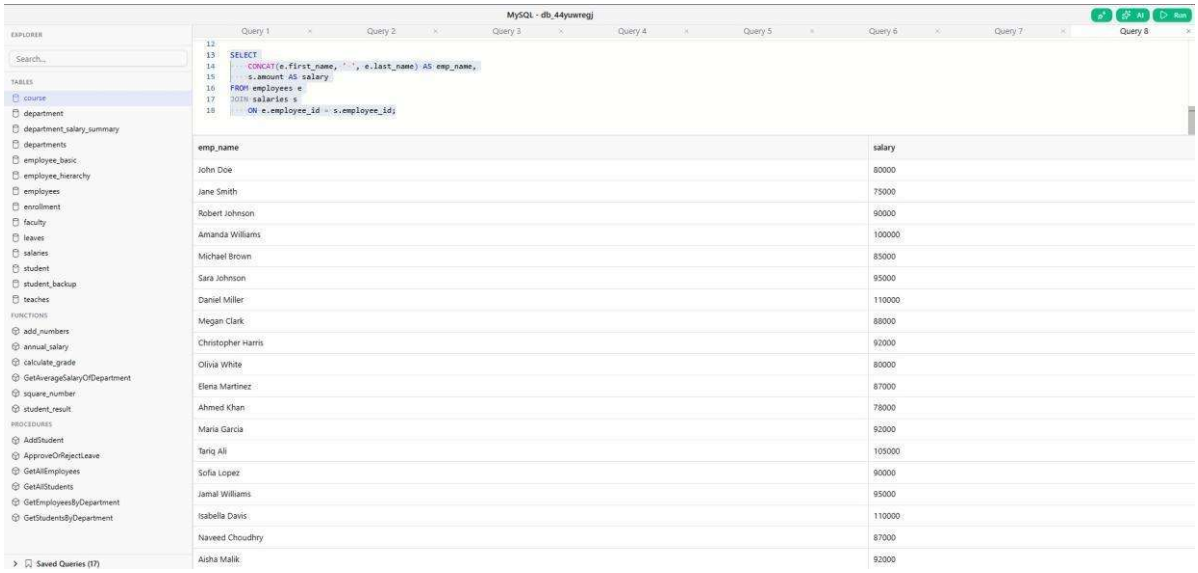
SELECTION

```
SELECT
  e.employee_id,
  e.first_name,
  e.last_name,
  s.amount AS salary
FROM employees e
JOIN salaries s
  ON e.employee_id = s.employee_id
WHERE s.amount > 80000;
```



PROJECTION

```
SELECT
  CONCAT(e.first_name, ' ', e.last_name) AS emp_name,
  s.amount AS salary
FROM employees e
JOIN salaries s
  ON e.employee_id = s.employee_id;
```



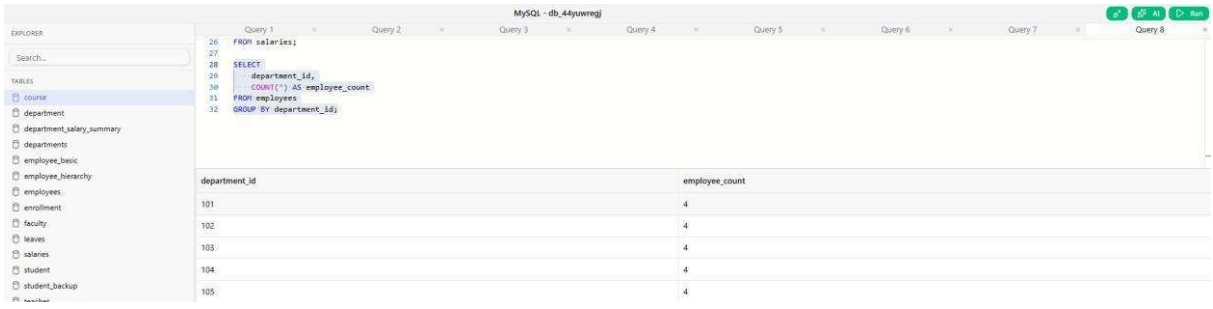
AGGREGATE FUNCTIONS

```
SELECT
   COUNT(DISTINCT employee_id) AS total_employees,
   SUM(amount) AS total_salary,
   AVG(amount) AS average_salary,
   MAX(amount) AS highest_salary,
   MIN(amount) AS lowest_salary
FROM salaries;
```



GROUP BY

```
SELECT
        dept_id,
        COUNT(*) AS employee_count
FROM Employee
GROUP BY dept_id;
```



GROUP BY WITH JOIN

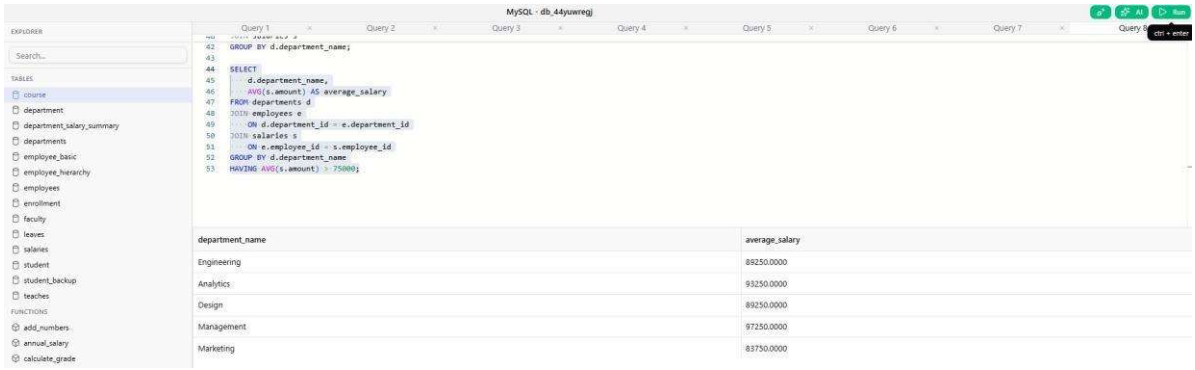
```
SELECT
   d.department_name,
   AVG(s.amount) AS average_salary
FROM departments d
JOIN employees e
   ON d.department_id = e.department_id
JOIN salaries s
   ON e.employee_id = s.employee_id
GROUP BY d.department_name;
```



department_name	average_salary
Engineering	89250.0000
Analytics	93250.0000
Design	89250.0000
Management	97250.0000
Marketing	83750.0000

HAVING

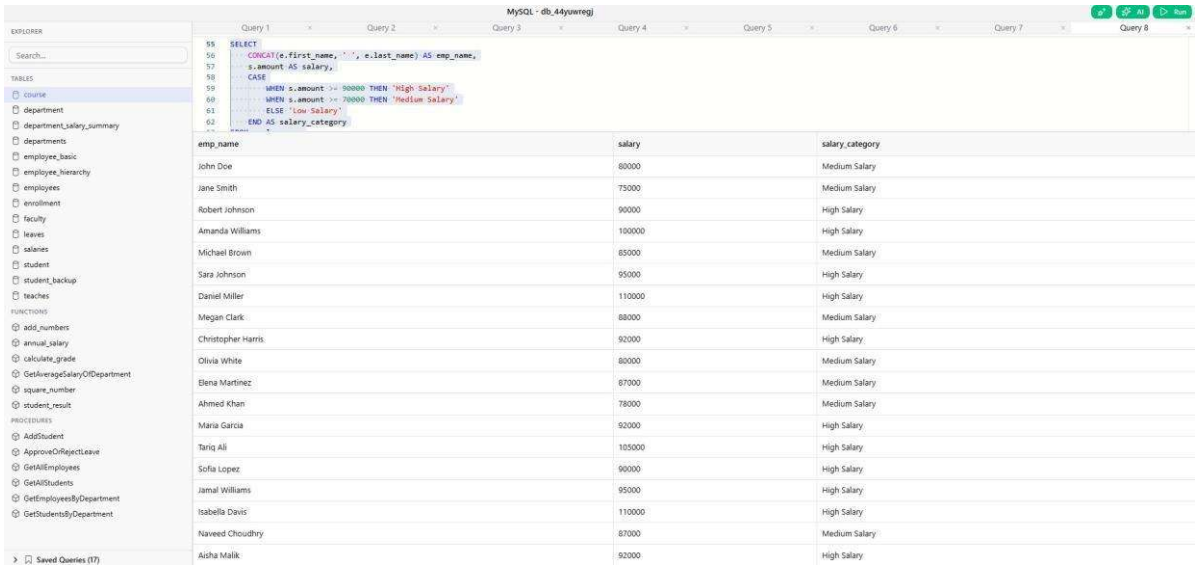
```
SELECT
   d.department_name,
   AVG(s.amount) AS average_salary
FROM departments d
JOIN employees e
   ON d.department_id = e.department_id
JOIN salaries s
   ON e.employee_id = s.employee_id
GROUP BY d.department_name
HAVING AVG(s.amount) > 75000;
```



department_name	average_salary
Engineering	89250.0000
Analytics	93250.0000
Design	89250.0000
Management	97250.0000
Marketing	83750.0000

CASE EXPRESSIONS

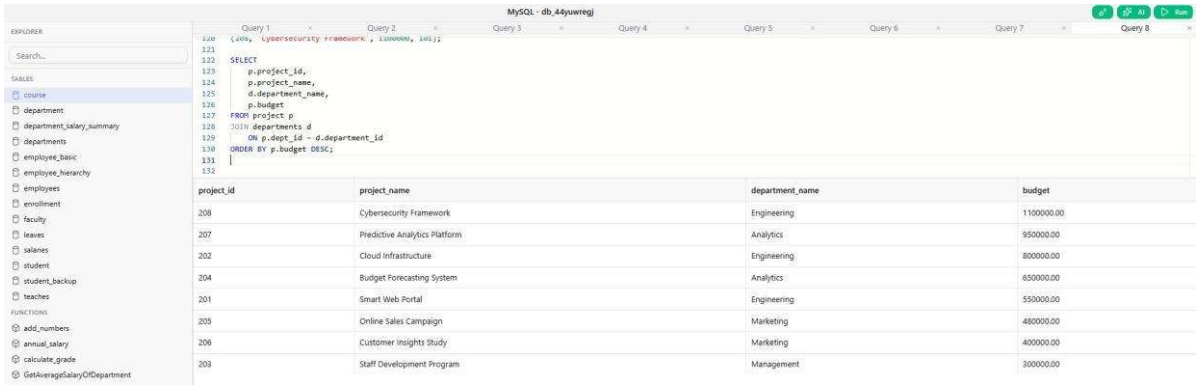
```
SELECT
  CONCAT(e.first_name, ' ', e.last_name) AS emp_name,
  s.amount AS salary,
  CASE
    WHEN s.amount >= 90000 THEN 'High Salary'
    WHEN s.amount >= 70000 THEN 'Medium Salary'
    ELSE 'Low Salary'
  END AS salary_category
FROM employees e
JOIN salaries s
  ON e.employee_id = s.employee_id;
```



emp_name	salary	salary_category
John Doe	80000	Medium Salary
Jane Smith	75000	Medium Salary
Robert Johnson	90000	High Salary
Amanda Williams	100000	High Salary
Michael Brown	85000	Medium Salary
Sara Johnson	95000	High Salary
Daniel Miller	110000	High Salary
Megan Clark	88000	Medium Salary
Christopher Harris	92000	High Salary
Olivia White	80000	Medium Salary
Elena Martinez	87000	Medium Salary
Ahmed Khan	78000	Medium Salary
Maria Garcia	92000	High Salary
Tariq Ali	105000	High Salary
Sofia Lopez	90000	High Salary
Jamal Williams	95000	High Salary
Isabella Davis	110000	High Salary
Naveed Choudhry	87000	Medium Salary
Aisha Malik	92000	High Salary

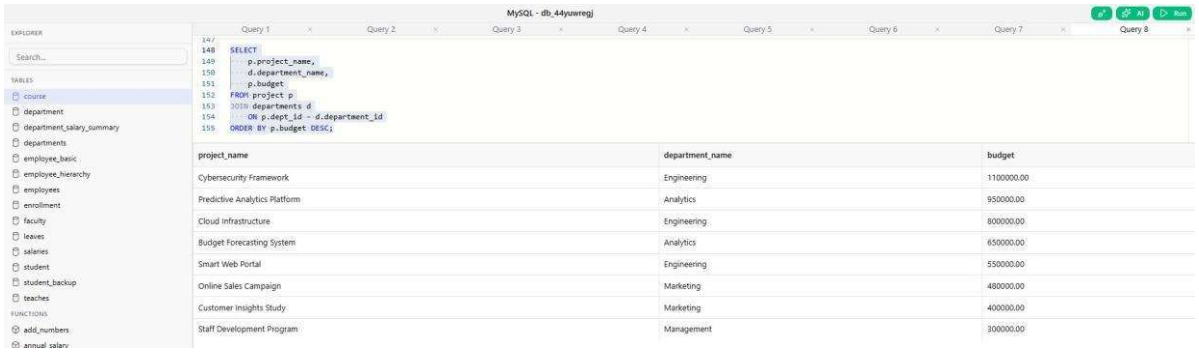
COMPLETE EMPLOYEE-DEPARTMENT REPORT

```
SELECT
  p.project_id,
  p.project_name,
  d.department_name,
  p.budget
FROM project p
JOIN departments d
  ON p.dept_id = d.department_id
ORDER BY p.budget DESC;
```



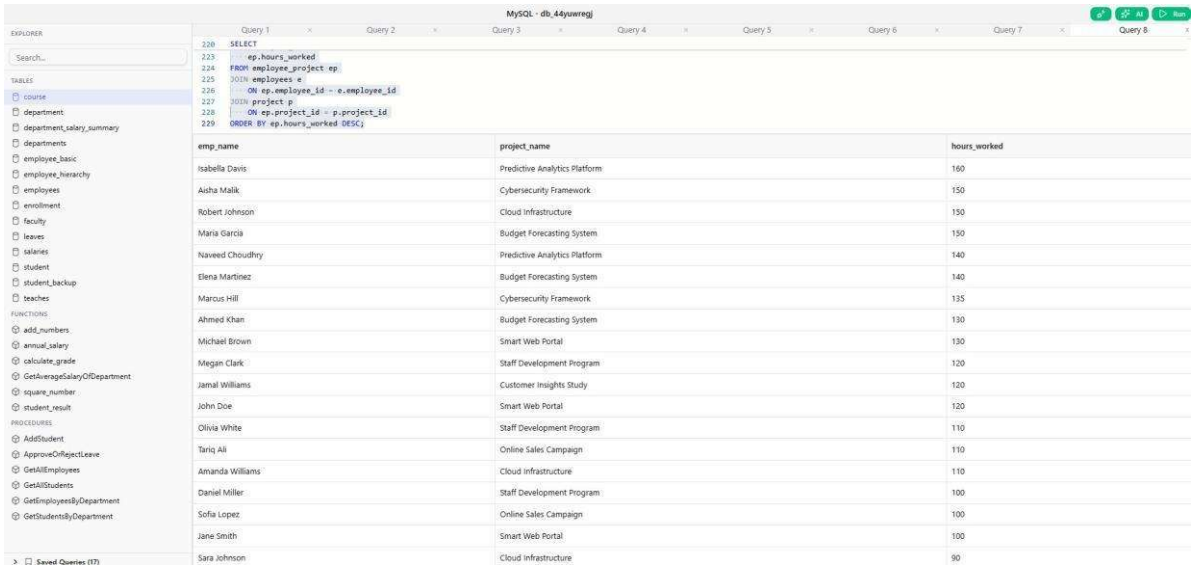
PROJECT DETAILS

```
SELECT
   p.project_name,
   d.department_name,
   p.budget
FROM project p
JOIN departments d
   ON p.dept_id = d.department_id
ORDER BY p.budget DESC;
```



EMPLOYEES WORKING ON PROJECTS

```
SELECT
   CONCAT(e.first_name, ' ', e.last_name) AS emp_name,
   p.project_name,
   ep.hours_worked
FROM employee_project ep
JOIN employees e
   ON ep.employee_id = e.employee_id
JOIN project p
   ON ep.project_id = p.project_id
ORDER BY ep.hours_worked DESC;
```



emp_name	project_name	hours_worked
Isabella Davis	Predictive Analytics Platform	160
Aisha Malik	Cybersecurity Framework	150
Robert Johnson	Cloud Infrastructure	150
Maria Garcia	Budget Forecasting System	150
Naveed Choudhry	Predictive Analytics Platform	140
Elena Martinez	Budget Forecasting System	140
Marcus Hill	Cybersecurity Framework	135
Ahmed Khan	Budget Forecasting System	130
Michael Brown	Smart Web Portal	130
Megan Clark	Staff Development Program	120
Jamal Williams	Customer Insights Study	120
John Doe	Smart Web Portal	120
Olivia White	Staff Development Program	110
Tariq Ali	Online Sales Campaign	110
Amanda Williams	Cloud Infrastructure	110
Daniel Miller	Staff Development Program	100
Sofia Lopez	Online Sales Campaign	100
Jane Smith	Smart Web Portal	100
Sara Johnson	Cloud Infrastructure	90